

Digital Professor: Focused Evaluation

Handwriting recognition and course-grounded tutoring

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What got done

Repeatable evaluation set

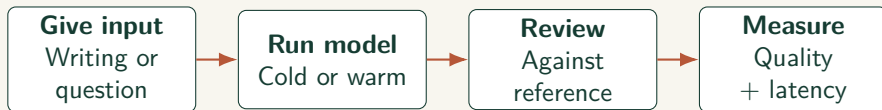
- ▶ 10 handwritten mathematics cases
- ▶ 8 course-grounded tutoring scenarios
- ▶ Cold and warm execution modes
- ▶ Saved outputs and review sheets

First measured baseline

- ▶ Recognition accuracy and failures
- ▶ Tutoring correctness and grounding
- ▶ Consistency and response latency
- ▶ 46 requests with no request errors

Outcome: a reusable baseline for measuring future improvements.

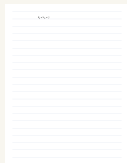
Evaluation workflow



- ▶ **Handwriting:** preserve mathematical meaning and the student's work.
- ▶ **Tutoring:** answer correctly from available course material and express uncertainty.
- ▶ **System:** remain consistent across repeats without excessive wait time.

Handwriting set

Case	Challenge
h01	Clean equation
h02	Fraction and exponents
h03	Subscripts and superscripts
h04	Matrix layout
h05	Multistep solution



h07

Case	Challenge
h06	Inequalities and constraints
h07	Ambiguous symbols
h08	Preserve an incorrect answer
h09	Multiple colors
h10	Variable-width calligraphy



h09



h10

Handwriting results

Meaning preserved

16/21

Mathematical equivalence

Exact text match

5/21

Character error rate: **0.246**

- ▶ Exact matching penalized harmless LaTeX formatting changes.
- ▶ h08 preserved the intentionally wrong derivative.
- ▶ Multicolor h09 succeeded in both runs.
- ▶ h06 added a hat to A .
- ▶ h07 omitted the final 1.
- ▶ h10 confused plus signs, commas, and adjacency.

Finding: ambiguous and variable-width strokes need confirmation.

New test case: equivalent notation and review flags

Same equation, different notation

Equivalent optimization notation

$$\min_x c^T x \iff \underset{x}{\text{minimize}} c^T x$$

Equivalent regression notation

$$\alpha + \beta x \iff \beta_0 + \beta_1 x_1$$

when $\alpha = \beta_0$, $\beta = \beta_1$, and $x = x_1$

Expected: equivalent after verifying the symbol mapping.

Send uncertain work to human review

Flag the result when:

- ▶ symbol confidence is low,
- ▶ recognition methods disagree,
- ▶ equivalence cannot be established, or
- ▶ a formula appears mathematically inconsistent.

Important: preserve the student's original writing and label a possible error; do not silently correct it.

Equivalence checking

Current: dictionary comparison with live updates after a mapping is verified. comparison, and numerical checks.

Exploring: symbolic simplification, expression-tree

How the tutoring scores were calculated

Each response was compared with the case's predefined expected points and its cited course source, then assigned a 0–2 score in each dimension.

Correctness

0: wrong or materially misleading **1:** partly correct **2:** correct on the expected points

Grounding

0: unsupported or invented source/page **1:** partly supported or weak citation **2:** every course-specific claim supported by loaded text and citation

Uncertainty handling

0: guesses through ambiguity or missing material **1:** partly acknowledges limits **2:** clearly states limits, asks for missing input, or answers directly when no uncertainty is needed

The reported value is the arithmetic mean of the 25 reviewed response scores in that dimension.

Tutoring results

Mean scores across 25 responses

Correctness	1.16 / 2
Grounding	0.88 / 2
Uncertainty	1.44 / 2

Only **3/25** earned full correctness and grounding.

Consistency

All 8 repeated scenarios stayed in the same broad answer family.

- ▶ Cited pages 23 and 29 when only five pages were loaded.
- ▶ Sometimes cited page 4 for content on page 5.
- ▶ Invented a task instead of clarifying an incomplete request.

Finding: consistent answers can still be unsupported or wrong.

Cold and warm latency

Modality	Cold anchor	Warm same case	Warm median
Handwriting	5.84 s	2.56 s, 4.28 s	3.56 s
Tutoring	11.45 s	7.79 s, 9.97 s, 9.08 s	6.60 s

- ▶ Measurements cover the complete HTTP request.
- ▶ Only one cold request was collected per modality.
- ▶ Warm medians mix inputs of different complexity.

Treat these as a preliminary baseline rather than a final comparison.

Improved syllabus parsing test

Batch result

28/34
82.35% successful

The parser was tested on course name, course number, and semester extraction across 34 syllabus cases.

What improved

Most cases returned a usable course title and recognized common number and term formats, including departmental prefixes and cross-listed courses.

Remaining failure patterns

- ▶ Generic department or school text selected as the course name
- ▶ Extra slash or parentheses in cross-listed results
- ▶ Course number or semester absent in otherwise successful rows
- ▶ Six cases still failed course-name extraction

Next check: score each field separately so partial extraction errors are visible instead of counted as one overall success.

What works and what is blocked

Working well

- ▶ Reliable evaluation execution
- ▶ Most handwriting retained meaning
- ▶ Student errors were preserved
- ▶ Repeated behavior was consistent

Current blockers

- ▶ Incomplete page-level source access
- ▶ Unsupported or inaccurate citations
- ▶ Ambiguous and stylized handwriting
- ▶ Weak clarification of missing input
- ▶ Cost rates were not configured

Next to-do: page-level source access



Example request

"What projected-gradient update is shown on page 23?"

Required behavior

Retrieve page 23 directly, answer from that page, and cite it only after verifying that the page was supplied to the model.

If retrieval fails, the tutor should state that the page is unavailable instead of reconstructing or inventing its contents.

Next milestone: improve, then rerun



- ▶ Reject citations to pages the model did not receive.
- ▶ Ask for objectives and constraints when a problem is missing.
- ▶ Keep h07 and h10 as handwriting regression cases.
- ▶ Repeat cold starts and configure rates for cost reporting.

Use the baseline to measure improvement